

Task 1: Install firewalld on your Linux system, start the service, and verify its status using systemctl and firewall-cmd commands.

Ans :-

status

Step 1: Install firewalld

```
sudo dnf install firewalld -y
```

Step 2: Start the service

```
sudo systemctl start firewalld
```

Step 3: Enable it at boot

```
sudo systemctl enable firewalld
```

Step 4: Check service status

```
sudo systemctl status firewalld
```

Expected output:

Active: active (running)

Step 5: Verify using firewall-cmd

```
sudo firewall-cmd --state
```

Expected output:

running

**Task 2: Add a new rule to firewalld to allow incoming HTTP (port 80) traffic, then use firewall-cmd to confirm the rule is active.

Hint: Use the --permanent and --reload options to make changes persistent.**

Ans :-

Add HTTP service permanently

```
sudo firewall-cmd --permanent --add-service=http
```

Expected output

success

Reload firewall

```
sudo firewall-cmd --reload
```

Expected output

success

Verify HTTP is allowed

```
sudo firewall-cmd --list-services
```

Example output

```
cockpit dhcpv6-client ssh http
```

or

```
sudo firewall-cmd --list-all
```

Output

public (active)

target: default

interfaces: ens160

services: cockpit dhcpv6-client http ssh

ports:

Task 3: Temporarily block all incoming traffic on port 8080 using firewalld, then test by attempting to access a simple web server (e.g., Python's http.server) running on that port.

Ans :-

Start a temporary Python web server

```
python3 -m http.server 8080
```

Open another terminal.

Check if the server is listening

```
ss -tuln | grep 8080
```

Example

```
LISTEN 0 5 0.0.0.0:8080
```

Ensure port 8080 is NOT allowed

If it was previously allowed, remove it temporarily:

```
sudo firewall-cmd --remove-port=8080/tcp
```

or reject traffic explicitly:

```
sudo firewall-cmd --add-rich-rule='rule family="ipv4" port protocol="tcp" port="8080" reject'
```

Verify:

```
sudo firewall-cmd --list-rich-rules
```

Expected output

```
rule family="ipv4" port port="8080" protocol="tcp" reject
```

Test access

From another machine:

```
curl http://<server-ip>:8080
```

Expected result

Connection refused

or

Connection timed out

depending on the rule.

Remove the temporary rule after testing:

```
sudo firewall-cmd --remove-rich-rule='rule family="ipv4" port protocol="tcp" port="8080" reject'
```

Task 4: List all currently active firewalld zones and display the services and ports allowed in the public zone.

Ans :-

List active zones

```
sudo firewall-cmd --get-active-zones
```

Example

public

interfaces: ens160

Display the public zone configuration

```
sudo firewall-cmd --zone=public --list-all
```

Example

public (active)

target: default

interfaces: ens160

services: cockpit dhcpv6-client ssh http

ports:

protocols:

masquerade: no

forward-ports:

source-ports:

rich rules:

Display only services:

```
sudo firewall-cmd --zone=public --list-services
```

Display only ports:

```
sudo firewall-cmd --zone=public --list-ports
```

Task 5: Remove a custom rule you added earlier (such as the HTTP port 80 rule) and verify that the change has taken effect using firewall-cmd.

Ans :-

Remove HTTP service permanently

```
sudo firewall-cmd --permanent --remove-service=http
```

Expected output

success

Reload firewalld

```
sudo firewall-cmd --reload
```

Verify removal

```
sudo firewall-cmd --list-services
```

Expected output

```
cockpit dhcpv6-client ssh
```

or

```
sudo firewall-cmd --list-all
```

You should no longer see:

http